

DATA ANALYTICS

PYTHON ASSIGNMENT 2 :DATA STRUCTURES- LIST, DICTIONARY, SET & CONDITIONAL STATEMENTS

SUBMISSION:03.10.2026

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Data Structures - List, Dictionary, Set & Conditional Statements

List (Creation, Modification and Access):

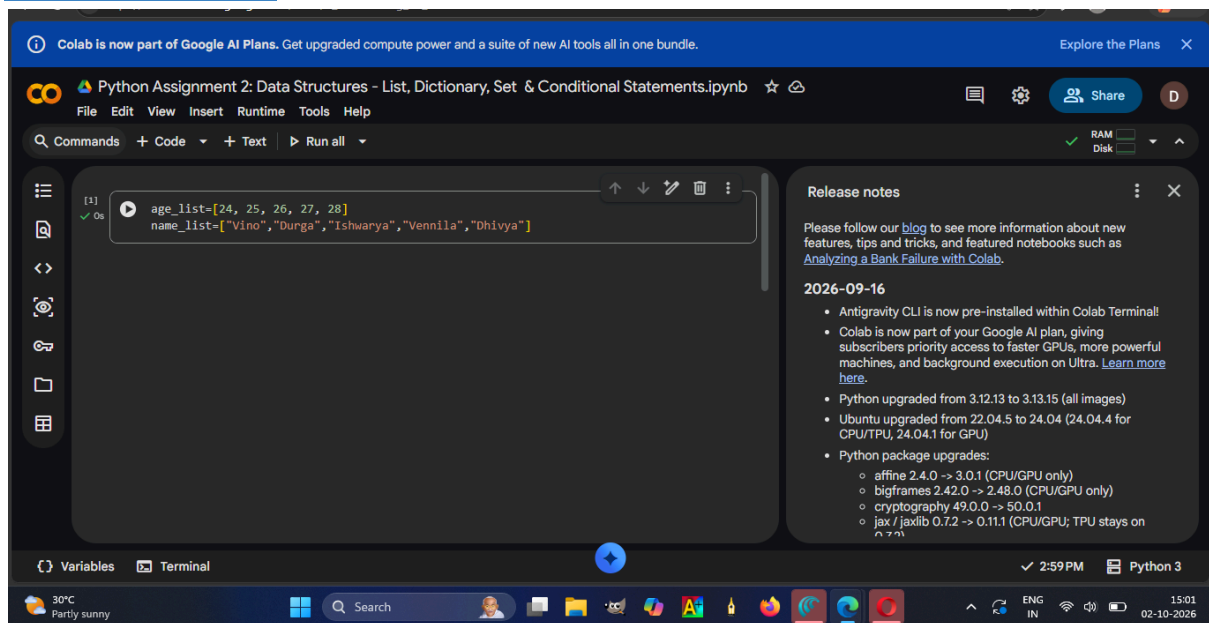
1. List Creation:

Task:

- Create a list named `age_list` with five integer elements. For eg., [24, 25, 26, 27, 28]
- Create a list named `name_list` with five string elements.

Solution :

Preview screenshot :



Explanation:

- First 1 had created the list with five integer elements and named the list as `age_list`.
- Then created the `name_list` with five string elements.

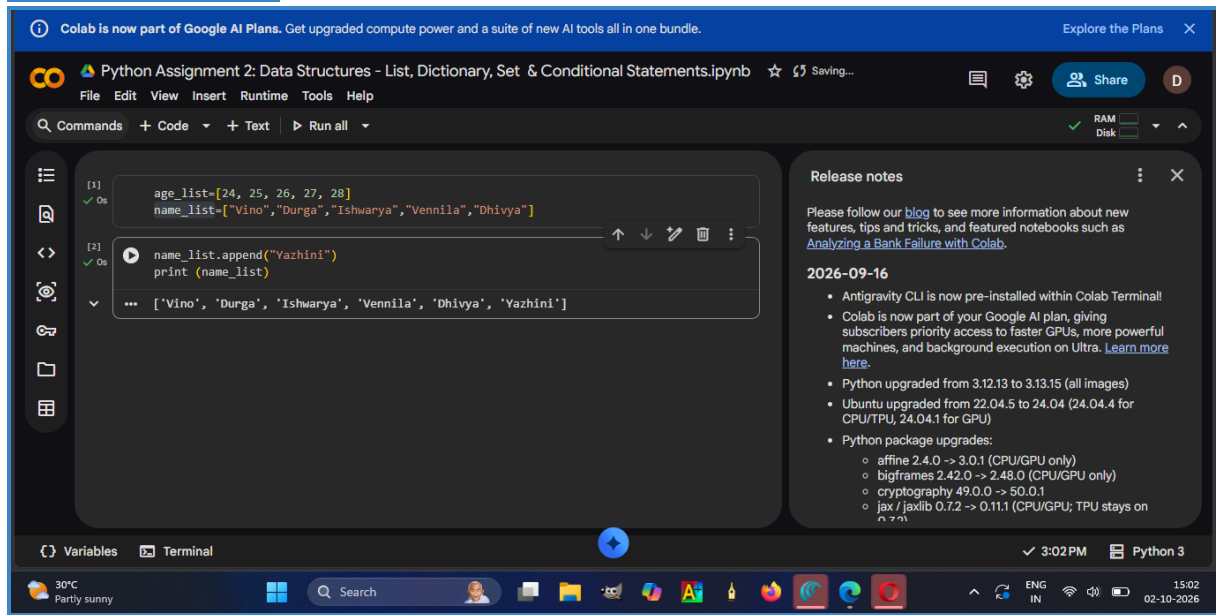
- The above mentioned screenshot represents the result.

2. List Operations / Modifications:

Task 1: Append the string "Yazhini" to name_list.

Solution :

Preview screenshot :



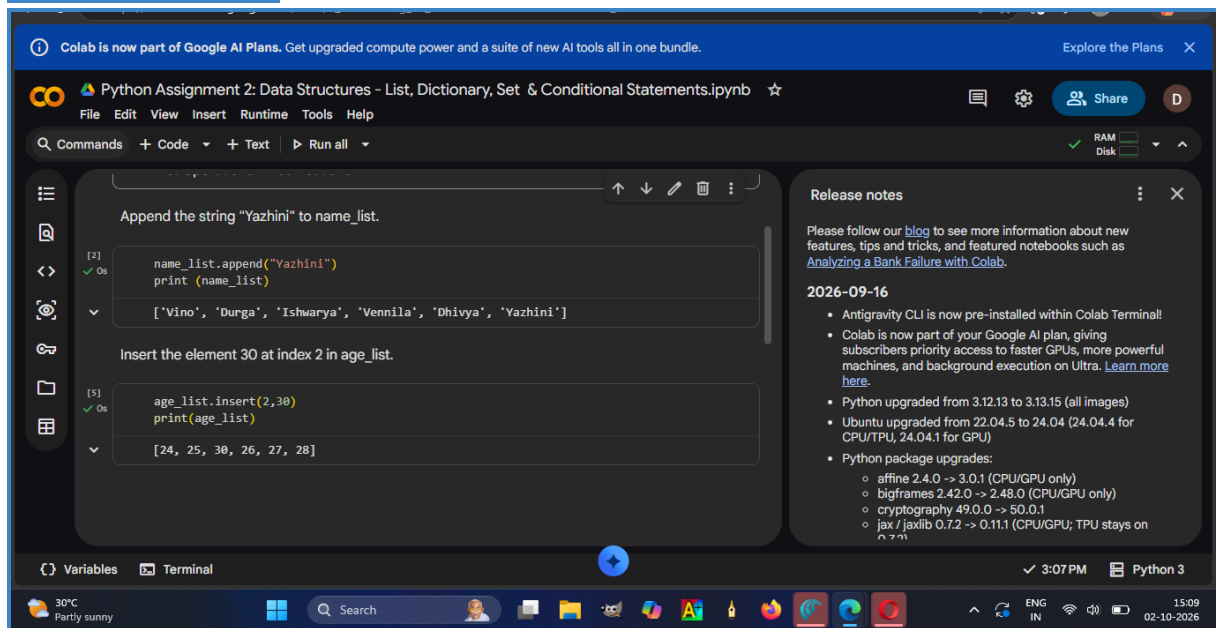
Explanation :

- I had used the append method for adding a string to name_list.
- The above mentioned screenshot represents the result.

Task 2 : Insert the element 30 at index 2 in age_list.

Solution:

Preview screenshot :



A screenshot of a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook shows two code cells. The first cell, labeled [2], contains the code to append the string "Yazhini" to a list named name_list and then print the list. The output shows the list: ['Vino', 'Durga', 'Ishwarya', 'Vennila', 'Dhivya', 'Yazhini']. The second cell, labeled [5], contains the code to insert the number 30 at index 2 of a list named age_list and then print the list. The output shows the list: [24, 25, 30, 26, 27, 28]. On the right side of the notebook, there is a "Release notes" panel for the date 2026-09-16, listing updates to the Colab environment.

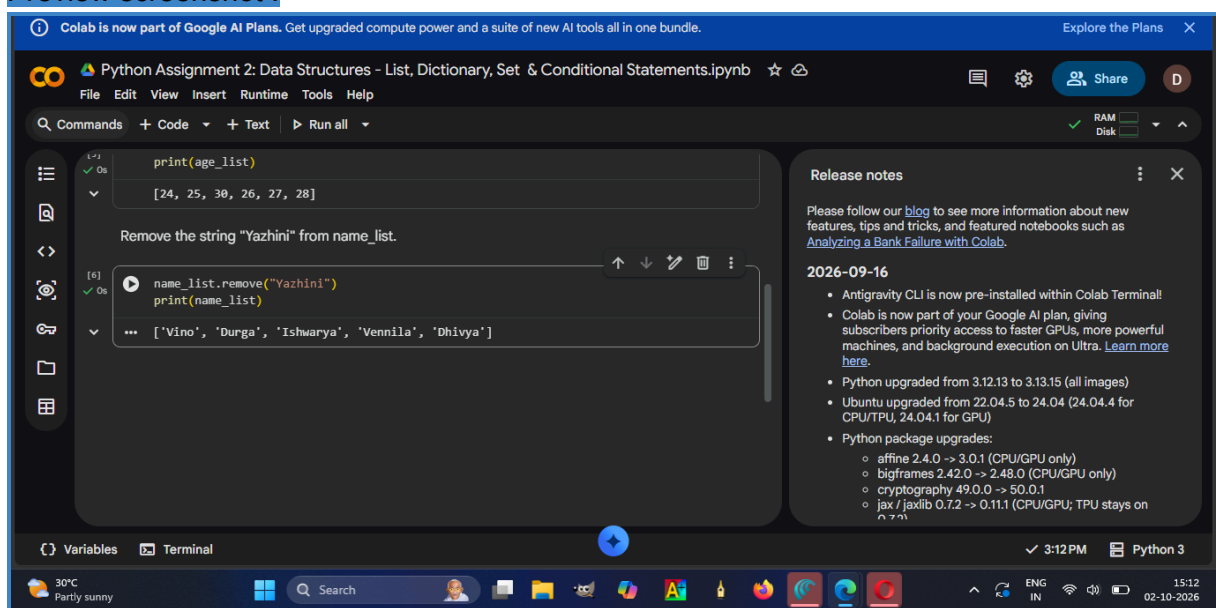
Explanation:

- I had used the insert method for inserting an element into age_list.
- The above mentioned screenshot represents the result.

Task 3: Remove the string "Yazhini" from name_list.

Solution:

Preview screenshot :



A screenshot of a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook shows two code cells. The first cell, labeled [3], contains the code to print the age_list, which outputs [24, 25, 30, 26, 27, 28]. The second cell, labeled [6], contains the code to remove the string "Yazhini" from the name_list and then print the list. The output shows the list: ['Vino', 'Durga', 'Ishwarya', 'Vennila', 'Dhivya']. On the right side of the notebook, there is a "Release notes" panel for the date 2026-09-16, listing updates to the Colab environment.

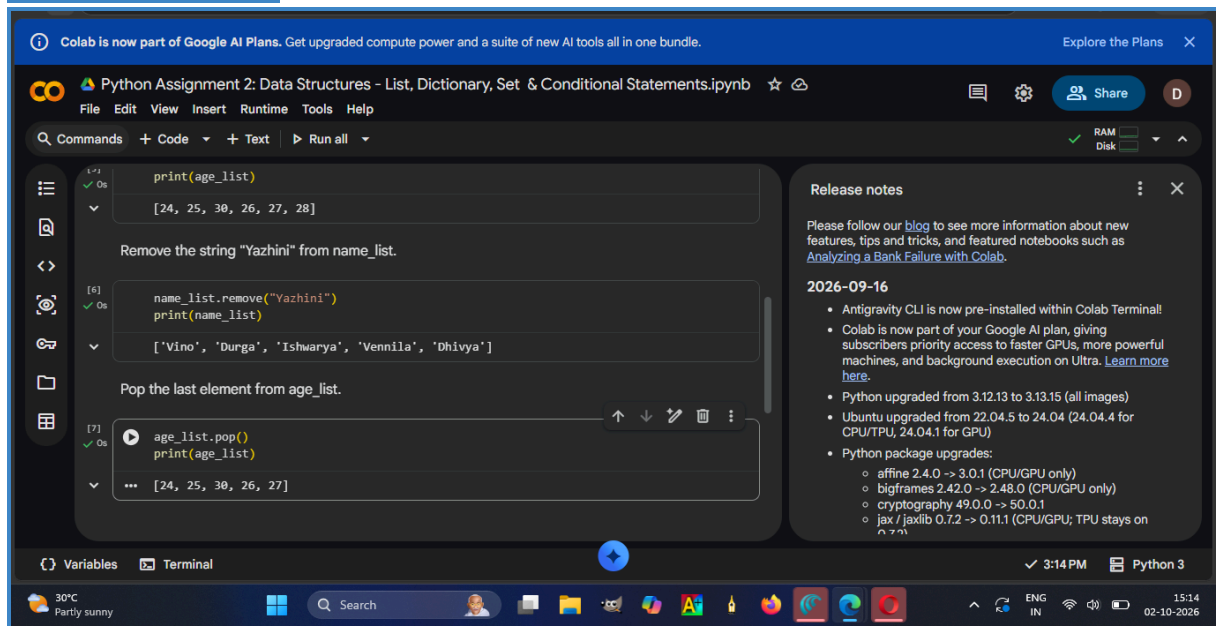
Explanation :

- I had used the insert method for inserting a string into name_list.
- The above mentioned screenshot represents the result.

Task 4: Pop the last element from age_list.

Solution:

Preview screenshot :



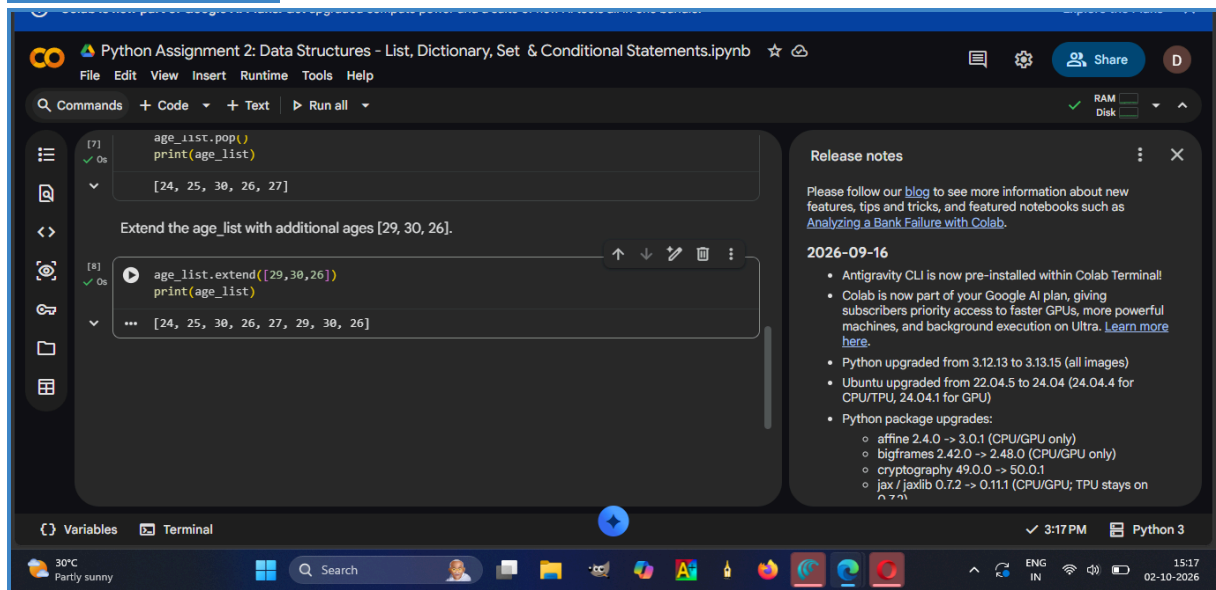
Explanation :

- I had used the pop method for deleting an element into age_list.
- The above mentioned screenshot represents the result.

Task 5 : Extend the age_list with additional ages [29, 30, 26].

Solution :

Preview screenshot :



A screenshot of a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook shows two code cells. The first cell, executed at 3:17 PM, contains the code `age_list.pop()` and `print(age_list)`, resulting in the output `[24, 25, 30, 26, 27]`. The second cell, also executed, contains the code `age_list.extend([29, 30, 26])` and `print(age_list)`, resulting in the output `[24, 25, 30, 26, 27, 29, 30, 26]`. A "Release notes" panel on the right lists updates from 2026-09-16, including Antigravity CLI, Colab's integration with Google AI, and various package upgrades. The bottom status bar shows the time as 3:17 PM and the Python version as Python 3.

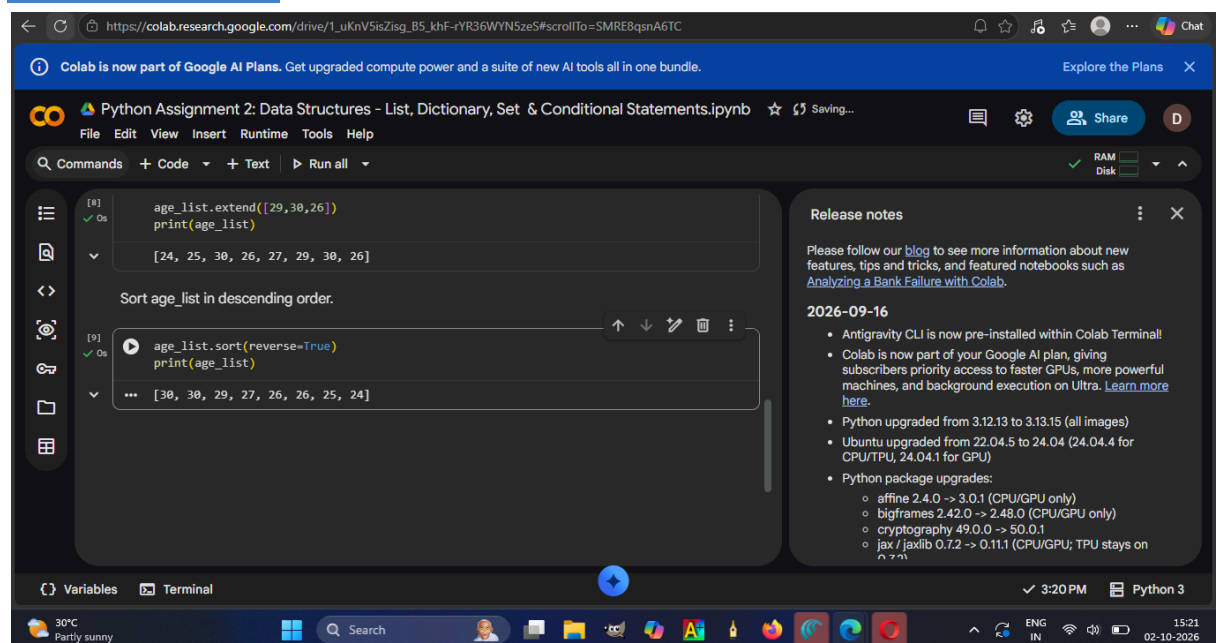
Explanation :

- I had used the extend method to add [29, 30, 26] into age_list.
- The above mentioned screenshot represents the result.

Task 6 : Sort age_list in descending order.

Solution :

Preview screenshot :



A screenshot of a Google Colab notebook showing the sorting of the age_list. The first code cell, executed at 3:20 PM, contains the code `age_list.extend([29, 30, 26])` and `print(age_list)`, resulting in the output `[24, 25, 30, 26, 27, 29, 30, 26]`. The second code cell, also executed, contains the code `age_list.sort(reverse=True)` and `print(age_list)`, resulting in the output `[30, 30, 29, 27, 26, 26, 25, 24]`. The "Release notes" panel on the right is identical to the previous screenshot. The bottom status bar shows the time as 3:20 PM and the Python version as Python 3.

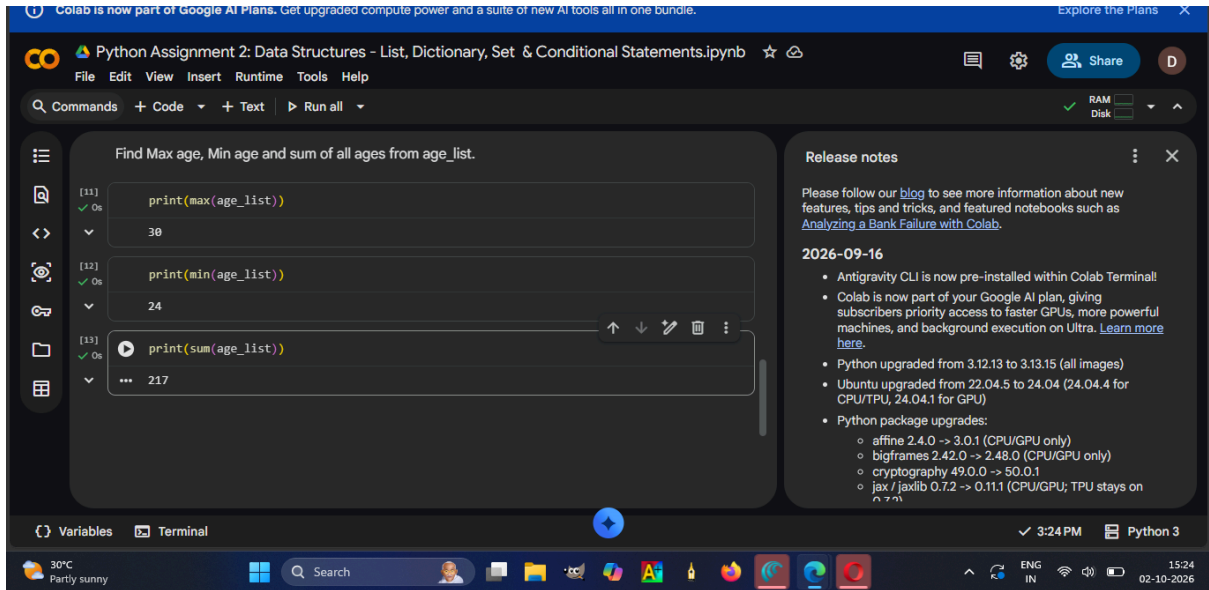
Explanation :

- I had used the sort method to sort age_list with reverse=true so it sorted the age list in descending order.
- The above mentioned screenshot represents the result.

Task 7: Find Max age, Min age and sum of all ages from age_list.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook contains three code cells, each with a "Run" button and a status indicator. The first cell contains `print(max(age_list))` and outputs `30`. The second cell contains `print(min(age_list))` and outputs `24`. The third cell contains `print(sum(age_list))` and outputs `217`. On the right side of the notebook, there is a "Release notes" panel for the date 2026-09-16, listing updates such as "Antigravity CLI is now pre-installed within Colab Terminal!", "Colab is now part of your Google AI plan", "Python upgraded from 3.12.13 to 3.13.15", and "Ubuntu upgraded from 22.04.5 to 24.04". The bottom of the screen shows a Windows taskbar with various application icons and system information like "30°C Partly sunny" and "15:24 02-10-2026".

Explanation :

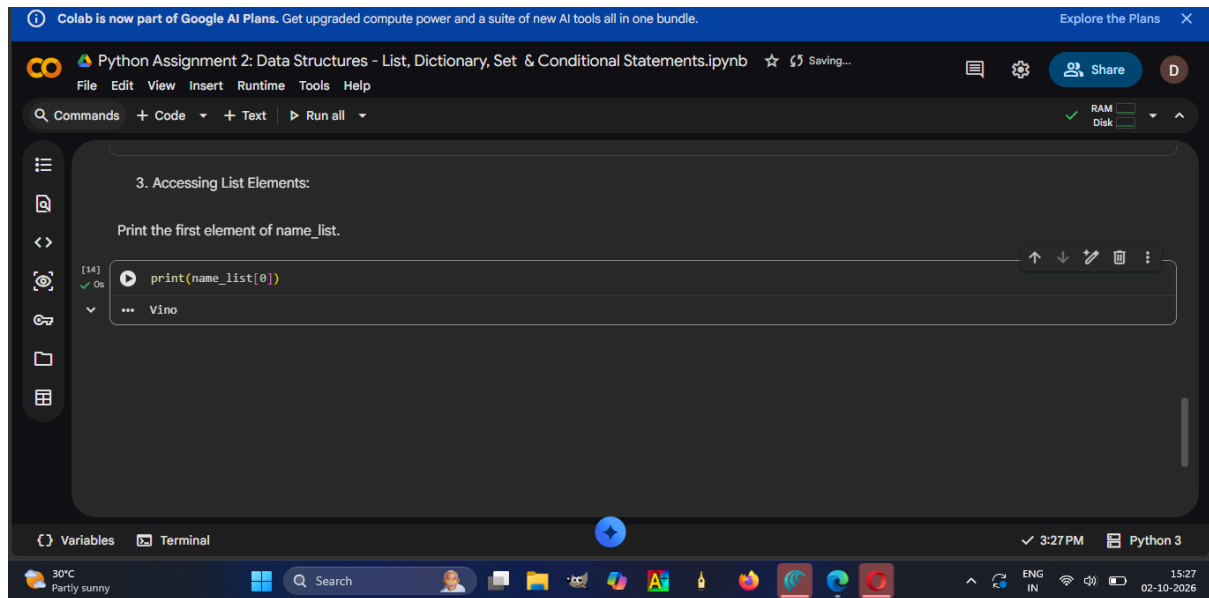
- I had used the max method to find the maximum age from the age_list. And a method to find minimum age from the age_list. And a sum method to find the sum of all ages from age_list.
- The above mentioned screenshot represents the result.

3. Accessing List Elements:

Task 1: Print the first element of name_list.

Solution :

Preview screenshot :



Explanation :

- I had used the indexing method for printing the first element in name_list.
- Here 0 represents the first element in the name_list.
- The above mentioned screenshot represents the result.

Task 2:Print the last element of name_list.

Solution :

Preview screenshot :

```
[14] print(name_list[0])
✓ Os
Vino

Print the last element of name_list.

[16] print(name_list[4])
✓ Os
Dhivya
```

Explanation :

- I had used the indexing method for printing the last element in name_list.
- Here 4 represents the last element in the name_list.
- The above mentioned screenshot represents the result.

Task 3 :Print the elements from index 2 to index 4 in name_list.

Solution :

Preview screenshot :

```
[16] print(name_list[4])
✓ Os
Dhivya

Print the elements from index 2 to index 4 in name_list.

[17] print(name_list[2:5])
✓ Os
["Ishwarya", "Vennila", "Dhivya"]
```

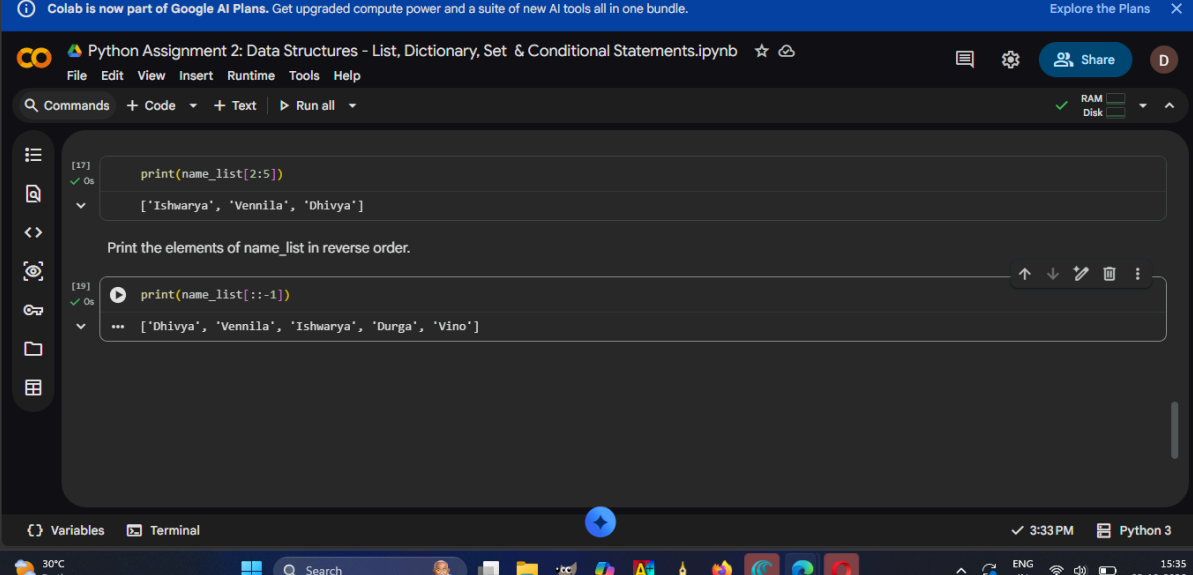
Explanation :

- I had used the slicing method for printing the 2 to 4th element in name_list.
- Here 2 represents the third element 5 represents the last element in the name_list.
- The above mentioned screenshot represents the result.

Task 4: Print the elements of name_list in reverse order.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". It contains two code cells. The first cell, labeled [17], contains the code `print(name_list[2:5])` and its output is `['Ishwarya', 'Vennila', 'Dhivya']`. The second cell, labeled [19], contains the code `print(name_list[::-1])` and its output is `['Dhivya', 'Vennila', 'Ishwarya', 'Durga', 'Vino']`. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for running code, and a status bar at the bottom showing system information like temperature and time.

```
[17]: print(name_list[2:5])
      ['Ishwarya', 'Vennila', 'Dhivya']

Print the elements of name_list in reverse order.

[19]: print(name_list[::-1])
      ... ['Dhivya', 'Vennila', 'Ishwarya', 'Durga', 'Vino']
```

Explanation :

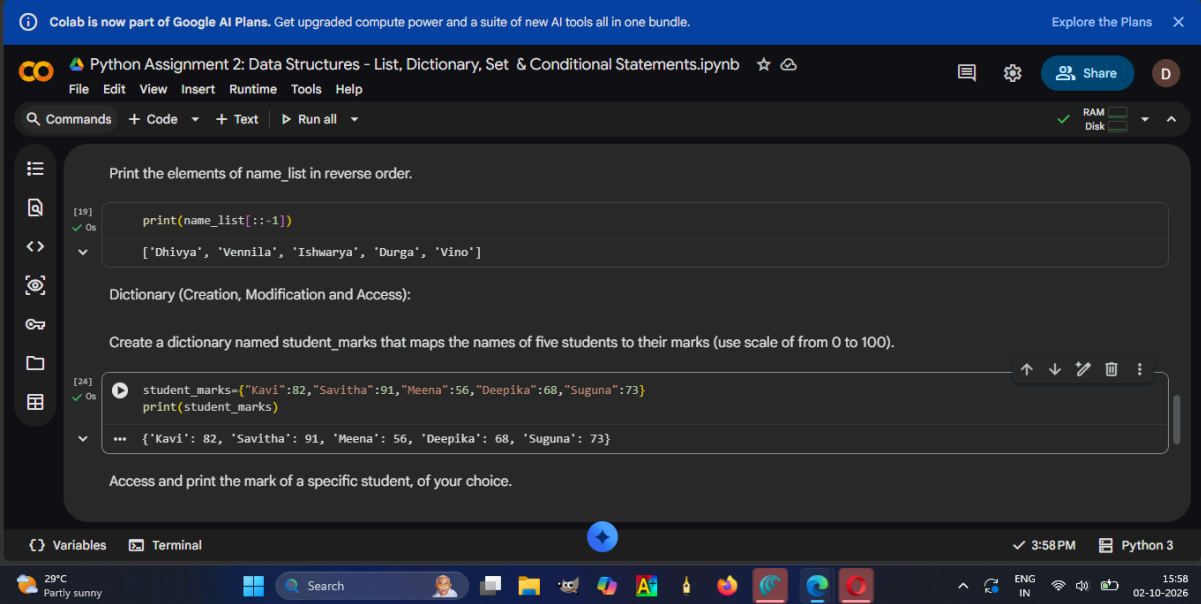
- I had used the slicing method for printing the name_list in reverse order.
- Here `::-1` is used to reverse the name_list.
- The above mentioned screenshot represents the result.

Dictionary (Creation, Modification and Access):

Task 1 : Create a dictionary named `student_marks` that maps the names of five students to their marks (use scale of from 0 to 100).

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook contains three code cells. The first cell prints the elements of a list in reverse order. The second cell creates a dictionary named `student_marks` and prints it. The third cell prints the mark of a specific student.

```
[19] print(name_list[::-1])
[19] ['Dhivya', 'Vennila', 'Ishwarya', 'Durga', 'Vino']

Dictionary (Creation, Modification and Access):

Create a dictionary named student_marks that maps the names of five students to their marks (use scale of from 0 to 100).

[24] student_marks={"Kavi":82,"Savitha":91,"Meena":56,"Deepika":68,"Suguna":73}
[24] print(student_marks)
[24] {'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73}

Access and print the mark of a specific student, of your choice.
```

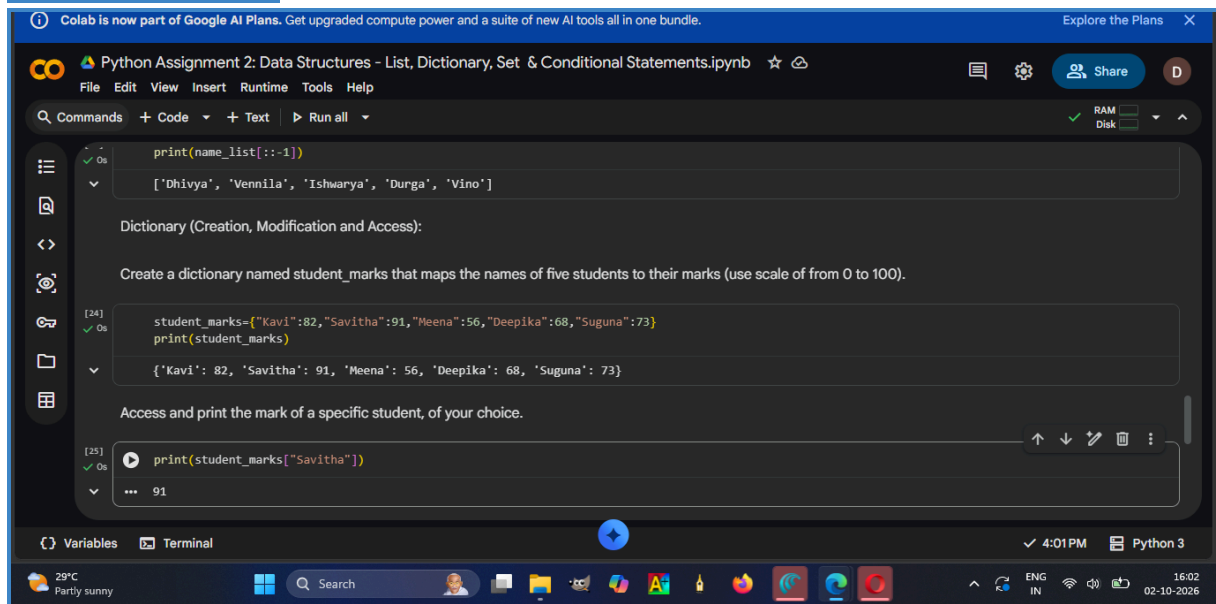
Explanation :

- I had created a dictionary named `student_marks`.
- Here key value pairs are separated by `' : '`.
- The above mentioned screenshot represents the result.

Task 2: Access and print the mark of a specific student, of your choice.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook contains the following code and output:

```
[24] print(name_list[::-1])
[24] ['Dhivya', 'Vennila', 'Ishwarya', 'Durga', 'Vino']

Dictionary (Creation, Modification and Access):

Create a dictionary named student_marks that maps the names of five students to their marks (use scale of from 0 to 100).

[24] student_marks={"Kavi":82,"Savitha":91,"Meena":56,"Deepika":68,"Suguna":73}
[24] print(student_marks)
[24] {'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73}

Access and print the mark of a specific student, of your choice.

[25] print(student_marks["Savitha"])
[25] 91
```

The bottom of the screenshot shows the Windows taskbar with the date 02-10-2026 and time 16:02.

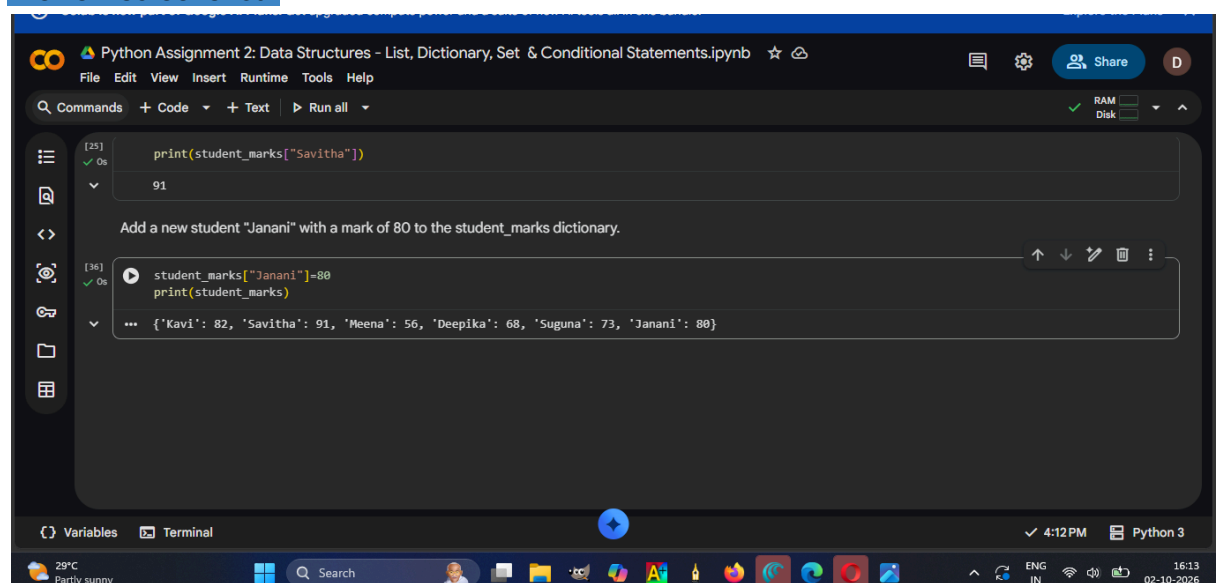
Explanation :

- I had used the key ("Savitha") to print the value (91) in student_marks.
- Here the student name is key and the mark is a value.
- The above mentioned screenshot represents the result.

Task 3 : Add a new student "Janani" with a mark of 80 to the student_marks dictionary.

Solution :

Preview screenshot :



The screenshot shows the same Google Colab notebook as before, but with additional code and output:

```
[25] print(student_marks["Savitha"])
[25] 91

Add a new student "Janani" with a mark of 80 to the student_marks dictionary.

[36] student_marks["Janani"]=80
[36] print(student_marks)
[36] {'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73, 'Janani': 80}
```

The bottom of the screenshot shows the Windows taskbar with the date 02-10-2026 and time 16:13.

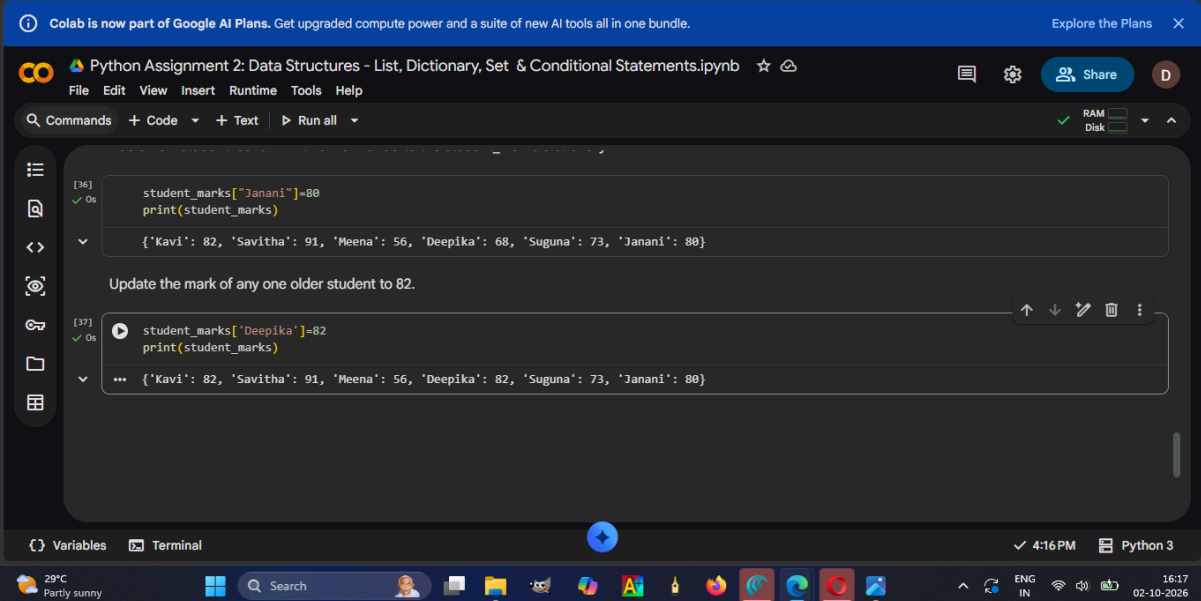
Explanation :

- I had added the student name “Janani” with mark 80 to student_marks.
- The above mentioned screenshot represents the result.

Task 4 : Update the mark of any one older student to 82.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". It contains two code cells. The first cell, labeled [36], sets `student_marks["Janani"] = 80` and prints the dictionary, resulting in `{'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73, 'Janani': 80}`. The second cell, labeled [37], updates `student_marks['Deepika'] = 82` and prints the dictionary, resulting in `{'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 82, 'Suguna': 73, 'Janani': 80}`. A text prompt "Update the mark of any one older student to 82." is visible between the two code cells. The interface includes a menu bar, a toolbar, and a status bar at the bottom showing the time as 4:16 PM and the environment as Python 3.

```
[36] student_marks["Janani"] = 80
      print(student_marks)

{'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73, 'Janani': 80}

Update the mark of any one older student to 82.

[37] student_marks['Deepika'] = 82
      print(student_marks)

... {'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 82, 'Suguna': 73, 'Janani': 80}
```

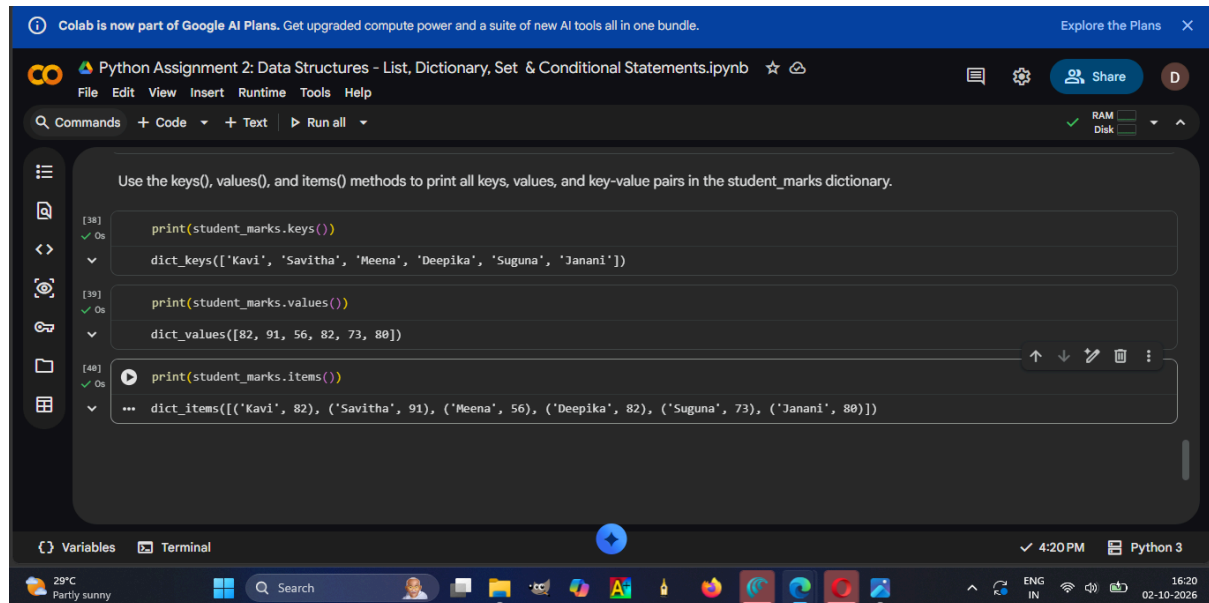
Explanation :

- I had updated the mark 82 to an old student name “Deepika” .
- The above mentioned screenshot represents the result.

Task 5: Use the `keys()`, `values()`, and `items()` methods to print all keys, values, and key-value pairs in the `student_marks` dictionary.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook contains three code cells. The first cell prints the keys of the `student_marks` dictionary, resulting in `dict_keys(['Kavi', 'Savitha', 'Meena', 'Deepika', 'Suguna', 'Janani'])`. The second cell prints the values, resulting in `dict_values([82, 91, 56, 82, 73, 80])`. The third cell prints the items, resulting in `dict_items([('Kavi', 82), ('Savitha', 91), ('Meena', 56), ('Deepika', 82), ('Suguna', 73), ('Janani', 80)])`. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar (Commands, + Code, + Text, Run all), and a status bar (Variables, Terminal, 4:20 PM, Python 3).

Explanation :

- I had used the `key` method to print only keys in the student mark dictionary .
- And I had used the `values` method to print only values in the student mark dictionary.
- And then I used the `item` method to print key- value pairs in the student mark dictionary.
- The above mentioned screenshot represents the result.

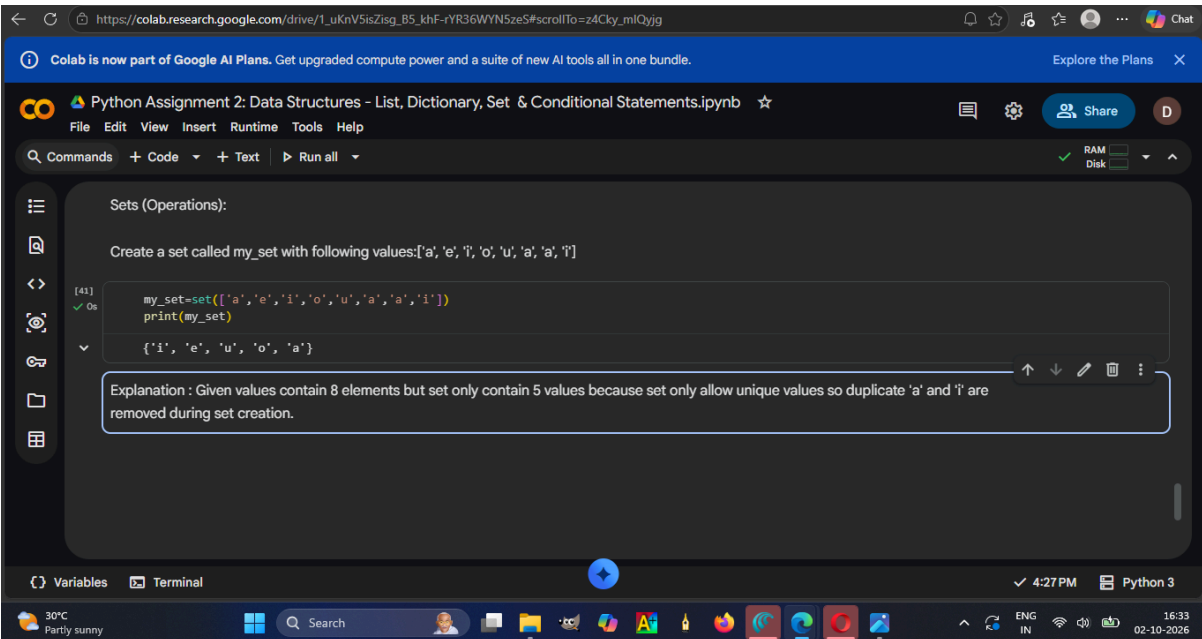
Sets (Operations):

Task 1 : Create a set called `my_set` with following values:

`['a','e','i','o','u','a','a','i']` . Analyse the output and provide explanation for the same.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook is open to a cell with the following code:

```
[41]: my_set=set(['a','e','i','o','u','a','a','i'])
      print(my_set)
```

The output of the cell is:

```
{'i', 'e', 'u', 'o', 'a'}
```

Below the output, there is an explanation:

Explanation : Given values contain 8 elements but set only contain 5 values because set only allow unique values so duplicate 'a' and 'i' are removed during set creation.

The notebook interface includes a menu bar with options like File, Edit, View, Insert, Runtime, Tools, and Help. There is also a "Share" button and a "Run all" button. The bottom status bar shows the time as 4:27 PM and the Python version as Python 3.

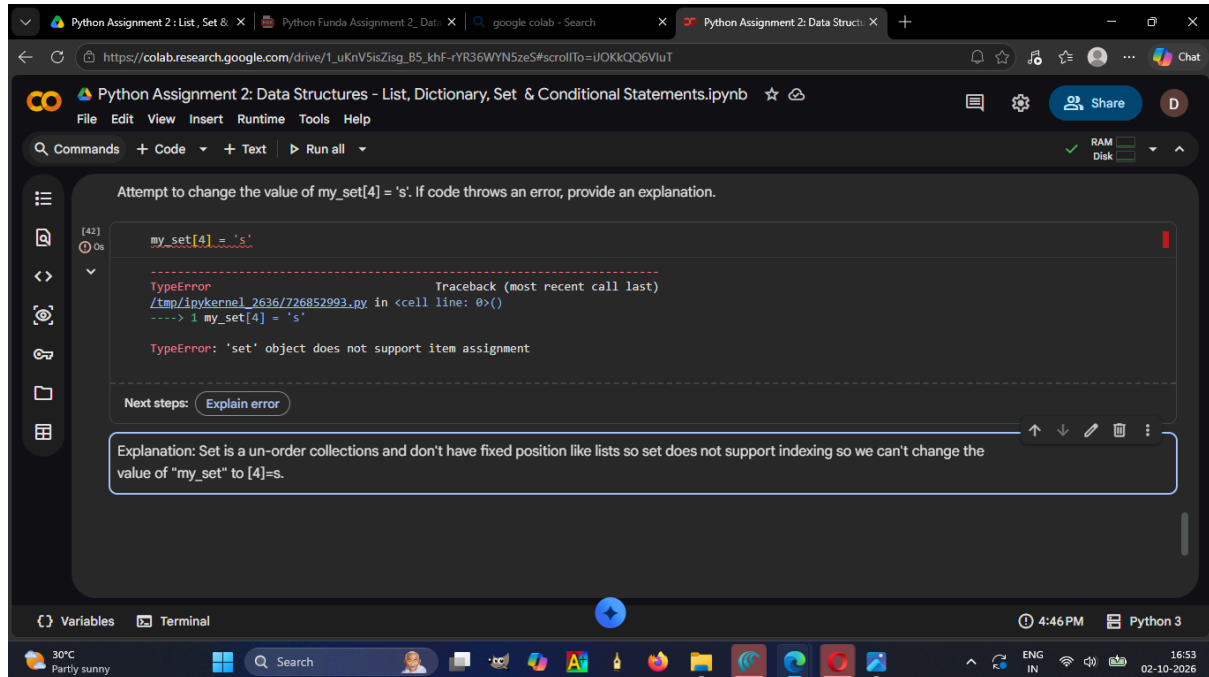
Explanation :

- I had created the set with a given value .
- Given values contain 8 elements but created sets only contain 5 values because sets only allow unique values so duplicates 'a' and 'i' are removed during set creation. .
- The above mentioned screenshot represents the result.

Task 2 : Attempt to change the value of `my_set[4] = 's'`. If code throws an error, provide an explanation.

Solution :

Preview screenshot :



```
[42] In [42]: my_set[4] = 's'
```

```
-----
```

```
TypeError                                 Traceback (most recent call last)
```

```
~/tmp/ipykernel_2636/726852993.py in <cell line: 0>()
```

```
----> 1 my_set[4] = 's'
```

```
TypeError: 'set' object does not support item assignment
```

Next steps: [Explain error](#)

Explanation: Set is a un-order collections and don't have fixed position like lists so set does not support indexing so we can't change the value of "my_set" to [4]=s.

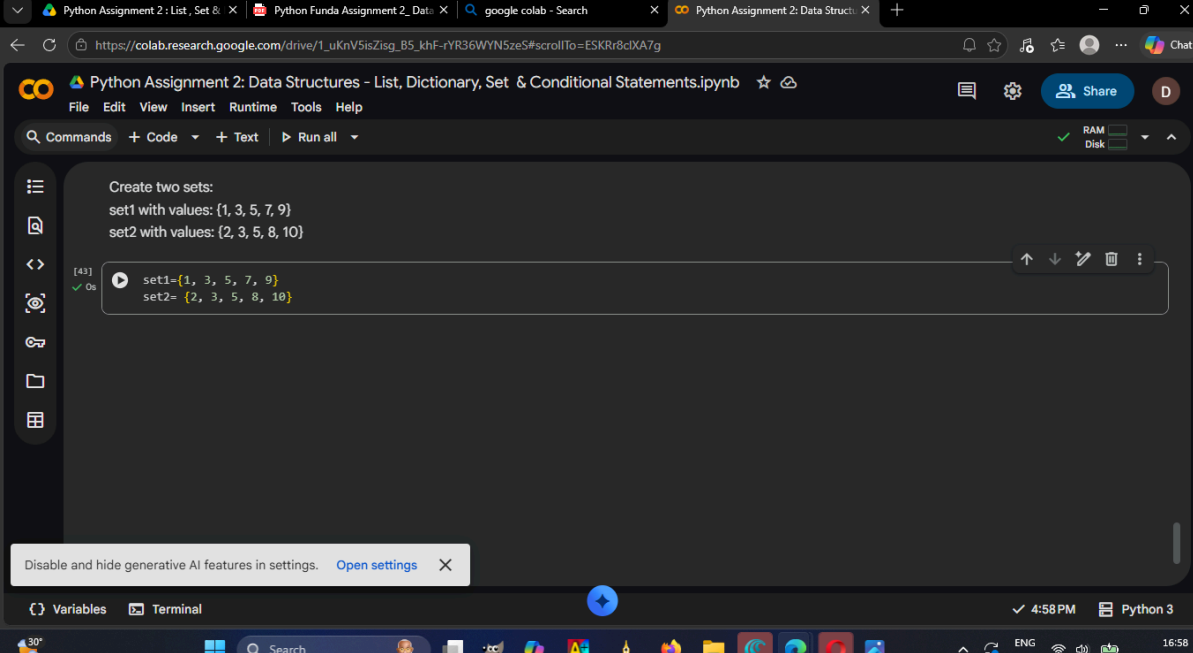
Explanation :

- I had attempted to change the value of `my_set[4]` but it showed an error because set is an un-order collection and doesn't have fixed position like lists so the set does not support indexing so we can't change the value of "my_set" to `[4]=s`.
- The above mentioned screenshot represents the result.

Task 3 : Create two sets: set1 with values: {1, 3, 5, 7, 9} , set2 with values: {2, 3, 5, 8, 10}

Solution:

Preview screenshot :



The screenshot shows a Google Colab notebook interface. The browser tabs at the top include 'Python Assignment 2 : List, Set & ...', 'Python Funda Assignment 2_ Data...', 'google colab - Search', and 'Python Assignment 2: Data Struct...'. The address bar shows a URL from colab.research.google.com. The notebook title is 'Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb'. The menu bar includes File, Edit, View, Insert, Runtime, Tools, and Help. The toolbar shows 'Commands', '+ Code', '+ Text', and 'Run all'. On the left sidebar, there are icons for file management and a 'RAM Disk' indicator. The main code cell contains the following text: 'Create two sets:', 'set1 with values: {1, 3, 5, 7, 9}', and 'set2 with values: {2, 3, 5, 8, 10}'. Below this, a code block shows the Python code:

```
set1={1, 3, 5, 7, 9}
set2= {2, 3, 5, 8, 10}
```

 The code is marked as '[43] ✓ 0s'. At the bottom of the notebook, there is a status bar with 'Variables', 'Terminal', a blue star icon, '4:58 PM', and 'Python 3'. The Windows taskbar at the very bottom shows the search bar, task view, and various application icons, with a system clock indicating 16:58 on 02-10-2026.

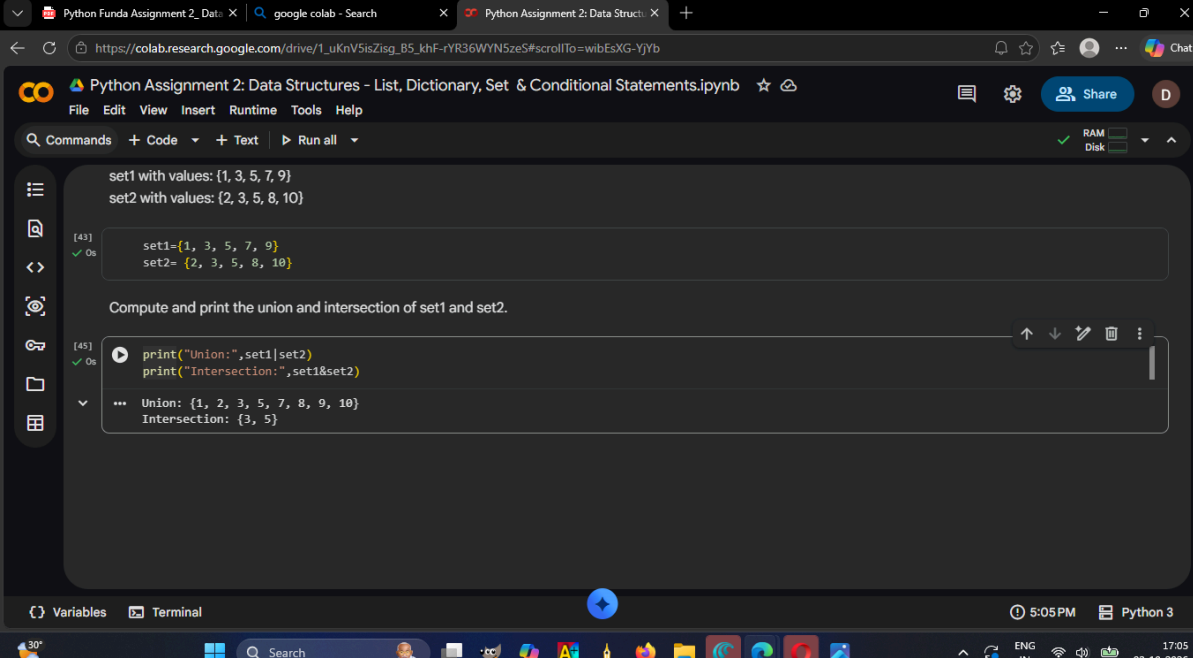
Explanation :

- I had created the two sets named set1 and set2 with the given values.
- The above mentioned screenshot represents the result.

Task 4 : Compute and print the union and intersection of set1 and set2.

Solution:

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook contains the following code:

```
set1 with values: {1, 3, 5, 7, 9}
set2 with values: {2, 3, 5, 8, 10}

set1={1, 3, 5, 7, 9}
set2={2, 3, 5, 8, 10}

Compute and print the union and intersection of set1 and set2.

print("Union:",set1|set2)
print("Intersection:",set1&set2)
```

The output of the code is displayed below the code cells:

```
Union: {1, 2, 3, 5, 7, 8, 9, 10}
Intersection: {3, 5}
```

Explanation :

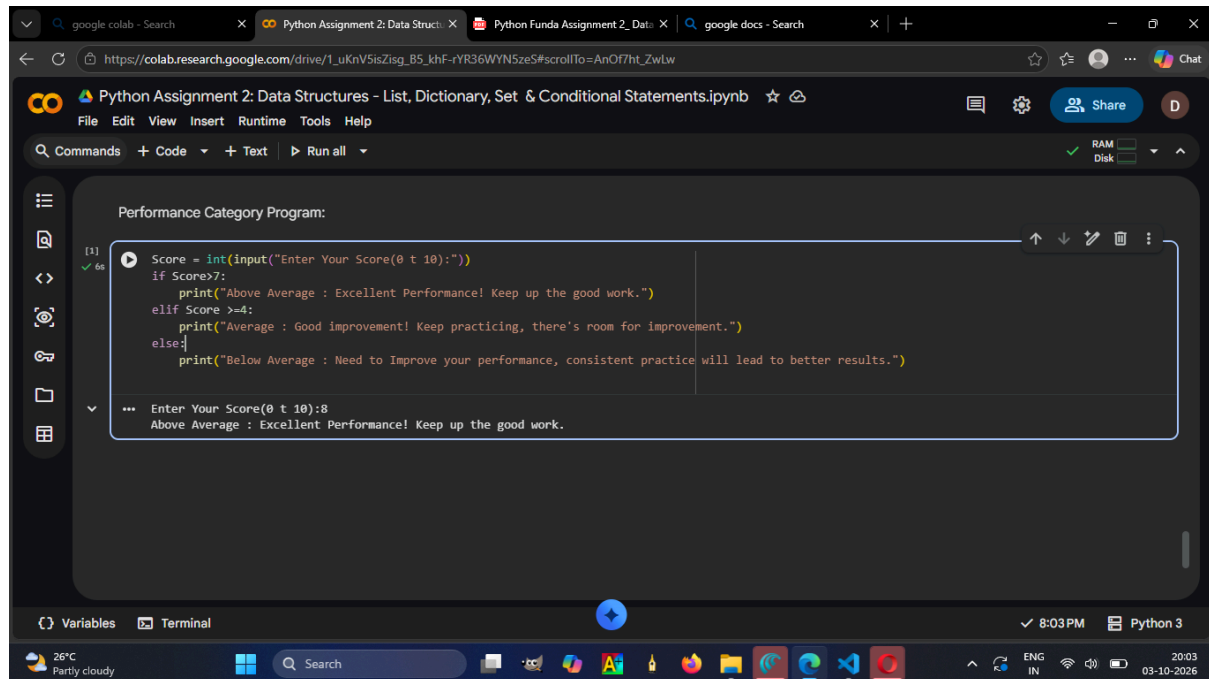
- I had used the “|” symbol for performing union and “&” symbol for performing intersection with set1 and set2 .
- The above mentioned screenshot represents the result.

Operators & Conditional Statements : (IF, ELIF, ELSE)

Task :Write Performance Category Program:

Solution :

Preview screenshot :



```
Performance Category Program:

Score = int(input("Enter Your Score(0 t 10):"))
if Score>7:
    print("Above Average : Excellent Performance! Keep up the good work.")
elif Score >=4:
    print("Average : Good improvement! Keep practicing, there's room for improvement.")
else:
    print("Below Average : Need to Improve your performance, consistent practice will lead to better results.")

... Enter Your Score(0 t 10):8
Above Average : Excellent Performance! Keep up the good work.
```

Explanation :

- I had created the nested conditional statement for the performance category program.
- First it will ask the input score with a message that “The score range should be from 0 to 10 : “.
- Then it will run the if and compare the input with a condition greater than 7 .
- If the score is greater then 7 the print the result as “Above Average : Excellent Performance! Keep up the good work.”
- If the score is not greater than 7, and it is greater than or equal to 4 ,it prints the result as "Average : Good improvement! Keep practicing, there's room for improvement."
- If the score is less than 4 then it will print the result as "Below Average : Need to Improve your performance, consistent practice will lead to better results."
- The above mentioned screenshot represents the result.